



PEDRO

GOMES PINTO

MASTER'S STUDENT IN FUNDAMENTAL MATHEMATICS

Exemplary diligence and strong adaptability. Equipped with solid skills in analysis, algebra, geometry, probability and topology, I am motivated by intellectual challenges and ready to apply my analytical abilities and rigor to innovative projects. A collaborative team member who thrives in group environments and values diverse perspectives.

EDUCATION

Master's Degree in Fundamental Mathematics (research) *Sept 2023 – Present*
Université de Lorraine, Metz

Bachelor's Degree in Mathematics and Applications *Graduated July 2023*
Université de Lorraine, Metz

Intensive Preparatory Program in Mathematics (CPGE) *Sept 2020 – July 2022*
Lycée Fabert, Metz

RESEARCH PROJECTS

Master's Thesis — Reilly-type inequalities in a variety of ambient manifolds *July 2026*
Université de Lorraine, Metz · Individual project

- Studied the orientability of smooth manifolds and De Rham cohomology.

Master's Thesis — Basic Theory of Manifolds *July 2025*
Université de Lorraine, Metz · Individual project

- Studied the orientability of smooth manifolds and De Rham cohomology.

Bachelor's Thesis — Knot Theory *May 2023*
Université de Lorraine, Metz · Collaborative project

- Studied topological invariants of knots and explored several applications.

EXPERIENCE

Part-Time Sales Associate *Apr 2022 – Aug 2024*
Carrefour City, Montigny-lès-Metz, France

- Managed product replenishment and inventory stocking.
- Led digital marketing and content creation for the store.
- Operated the point-of-sale (POS) system, processing customer transactions with accuracy and efficiency.
- Wrote an 18-page report on what the job consists of

Research Internship — First Eigenvalue of the Laplace-Beltrami Operator *Apr 2026 – July 2026*
Université de Lorraine, Metz, France

- Investigating spectral properties of the Laplace-Beltrami operator, focusing on bounds and behavior of the first eigenvalue in specific ambient spaces.
- Studying how geometric constraints (curvature, volume, boundary conditions) influence eigenvalue estimates.

CONTACT

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Fontoy, France

SKILLS

- Active listener
- Receptive to feedback
- Thrives in fast-paced environments
- Excellent written and verbal communication
- Strong analytical and problem-solving skills
- Curiosity in and outside of classes

LANGUAGES

- Portuguese — Native
- French — Fluent
- English — Fluent
- German — Intermediate
- Spanish — Intermediate
- Luxembourgish — Beginner
- Italian — Beginner